

Understanding Sampling Error

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Contents

1	Introduction	1
2	The Toy Population	2
3	Random Sampling and Sampling Distribution	3
4	Understanding P-values	5
5	Why Sample Size Matters	7
6	When Random Sampling Fails	8
A	Complete Sample List	11
B	Key Formulas	12
C	Proof of Unbiasedness	12
D	Concept Glossary	13

1 Introduction

1.1 The Central Challenge

Statistics faces a fundamental challenge: **we want to know the truth about everyone, but we can only observe data from some people.** This is the central paradox of statistical inference. How can we responsibly make claims about a population when we only have information from a sample?

In this tutorial, we'll explore what random sampling can guarantee—and what it cannot guarantee. You'll see that sampling error is unavoidable, but it can be understood, quantified, and incorporated into our reasoning.

1.2 The Asymmetric Relationship Between Population and Sample

Statistical inference rests on a fundamental asymmetry: researchers observe samples but make claims about populations. Let's make this concrete with an example.

Suppose the UBC Student Society (AMS) requests permanent funding for a foodbank from the university. The university has established a policy rule: funding will be approved only if students worry about their next meal at least 40% of the time on average. In other words, if the population mean $\mu \geq 0.40$, funding is provided.

The challenge: we need to estimate μ , but resource constraints limit us to surveying only 2 students. This simple scenario will help us understand the entire logic of statistical inference.

2 The Toy Population

2.1 Population Setup

To make the concept of sampling distribution completely transparent and computable, we'll work with a very small toy population. Imagine UBC has only 7 undergraduate students. We ask each student: "What fraction of time in the past week were you worried about your next meal?" Their answers are proportions between 0 and 1.

Here's our complete population:

Student ID	1	2	3	4	5	6	7
Worry fraction Y_i	0	0	0	0.5	0.5	1	1

Table 1: Toy population: Food insecurity data for 7 students

Students 1, 2, and 3 never worry about food ($Y = 0$). Students 4 and 5 worry half the time ($Y = 0.5$). Students 6 and 7 always worry ($Y = 1$).

2.2 Calculating Population Parameters

Let's calculate the population mean μ . This is the **truth** we want to know:

$$\mu = \frac{1}{N} \sum_{i=1}^N Y_i = \frac{0 + 0 + 0 + 0.5 + 0.5 + 1 + 1}{7} = \frac{3}{7} \approx 0.4286 \quad (1)$$

Notice that $\mu \approx 0.43 > 0.40$. According to the university's rule, food insecurity exceeds the 40% threshold, so **the truth warrants funding**. Keep this in mind—we'll see later that even when the truth is clear, samples can mislead us.

We can also calculate the population variance (using the finite population variance formula with $N - 1$ denominator):

$$S^2 = \frac{1}{N - 1} \sum_{i=1}^N (Y_i - \mu)^2 = \frac{1}{6} \sum_{i=1}^7 (Y_i - 3/7)^2 = \frac{17}{84} \approx 0.2024 \quad (2)$$

This variance measures variability among individuals in the population. It will play a role when we calculate standard errors later.

Key point: μ and S^2 are population parameters, denoted by Greek letters. They are fixed (though unknown in practice) characteristics of the entire population.

3 Random Sampling and Sampling Distribution

3.1 What is Simple Random Sampling

Due to resource constraints, we can only survey $n = 2$ students. We use **Simple Random Sampling Without Replacement (SRSWOR)**. What does this mean? Every possible 2-person combination has an equal probability of being selected.

How many ways can we choose 2 people from 7? We use the combination formula:

$$\binom{7}{2} = \frac{7!}{2!(7-2)!} = \frac{7 \times 6}{2 \times 1} = 21 \quad (3)$$

So there are **21 possible samples**, each with probability $1/21$. In actual research, we would only observe *one* of these 21 samples, but in principle it could be any one of them.

3.2 The Sample Mean

For any sample, we calculate the sample mean $\bar{Y}$:

$$\bar{Y} = \frac{1}{n} \sum_{j=1}^n Y_{(j)} = \frac{1}{2}(Y_{(1)} + Y_{(2)}) \quad (4)$$

Here $Y_{(j)}$ denotes the j -th observation in the sample.

Let's look at a few examples:

- **Sample** $\{1, 2\}$: values $(0, 0)$, sample mean $\bar{Y} = 0$
- **Sample** $\{1, 4\}$: values $(0, 0.5)$, sample mean $\bar{Y} = 0.25$
- **Sample** $\{4, 5\}$: values $(0.5, 0.5)$, sample mean $\bar{Y} = 0.5$
- **Sample** $\{6, 7\}$: values $(1, 1)$, sample mean $\bar{Y} = 1.0$

Notice that different samples produce different values of $\bar{Y}$. This variation—the difference in sample statistics across samples—is the essence of **sampling variability**.

Sample mean $\bar{Y}$	0.00	0.25	0.50	0.75	1.00
Number of samples	3	6	7	4	1
Probability	3/21	6/21	7/21	4/21	1/21
Percentage	14.3%	28.6%	33.3%	19.0%	4.8%

Table 2: Sampling distribution: All 21 possible sample means

3.3 Building the Complete Sampling Distribution

Now let's do something impossible in actual research: enumerate all 21 possible samples and calculate $\bar{Y}$ for each one. This exercise, while tedious, will make the concept of sampling distribution completely clear.

The complete list is in Appendix A. Here's the summary:

This distribution—the frequency distribution of all possible sample means—is the **sampling distribution** of $\bar{Y}$. It is the conceptual foundation of statistical inference.

If you visualize this as a histogram with $\bar{Y}$ values on the x-axis and frequency on the y-axis, you'll see that the most common value is $\bar{Y} = 0.50$ (occurring in 7 out of 21 samples), while extreme values like $\bar{Y} = 1.00$ are rare (occurring in only 1 out of 21 samples).

3.4 Three Key Properties of the Sampling Distribution

Carefully examining this sampling distribution reveals three crucial properties:

Property 1: Unbiasedness

The expected value of $\bar{Y}$ across all possible samples equals the true population mean:

$$\mathbb{E}[\bar{Y}] = \sum \bar{Y} \times P(\bar{Y}) \quad (5)$$

$$= 0.00 \times \frac{3}{21} + 0.25 \times \frac{6}{21} + 0.50 \times \frac{7}{21} + 0.75 \times \frac{4}{21} + 1.00 \times \frac{1}{21} \quad (6)$$

$$= \frac{0 + 1.5 + 3.5 + 3.0 + 1.0}{21} \quad (7)$$

$$= \frac{9}{21} = \frac{3}{7} = \mu \quad \checkmark \quad (8)$$

This means the sample mean is **unbiased**: on average, across all possible samples, it equals the true population mean. However—and this is crucial—*any individual sample* may deviate substantially from μ . Unbiasedness is a long-run property, not a guarantee for individual samples.

Property 2: Sampling variability

The variance of $\bar{Y}$ across all possible samples can be calculated:

$$\text{Var}(\bar{Y}) = \mathbb{E}[(\bar{Y} - \mu)^2] \quad (9)$$

$$= \sum (\bar{Y} - \mu)^2 \times P(\bar{Y}) \quad (10)$$

Working through the calculation (details omitted for brevity), we get:

$$\text{Var}(\bar{Y}) = 0.1012 \quad (11)$$

The square root of this variance is the **standard error**:

$$\text{SE}(\bar{Y}) = \sqrt{0.1012} \approx 0.318 \quad (12)$$

This number quantifies the typical distance between a sample mean and the true population mean.

For finite population sampling, there's a formula that relates this variance to population variance:

$$\text{Var}(\bar{Y}) = \frac{S^2}{n} \left(1 - \frac{n}{N}\right) \quad (13)$$

The factor $\left(1 - \frac{n}{N}\right)$ is called the **finite population correction (FPC)**. In our case:

$$\text{Var}(\bar{Y}) = \frac{0.2024}{2} \left(1 - \frac{2}{7}\right) = 0.1012 \times \frac{5}{7} \approx 0.0723 \quad (14)$$

Wait—this doesn't match! Why? Because when the sample is a substantial fraction of the population (here, $2/7 \approx 28.6\%$), the finite population correction matters. The formula above is the theoretical prediction; our enumeration gave us the exact answer.

Property 3: Predictable error range

Using the standard error, we can construct a **margin of error**. For approximately 95% coverage (using the normal approximation):

$$\text{MOE}_{95\%} = 1.96 \times \text{SE}(\bar{Y}) = 1.96 \times 0.318 \approx 0.623 \quad (15)$$

This means: in 95% of possible samples, the sample mean will fall within ± 0.623 of the true population mean $\mu = 0.43$.

Notice how large this margin is relative to the threshold of 0.40! This illustrates a fundamental point: **small samples have large uncertainty**.

4 Understanding P-values

4.1 The Research Question

Recall the university's funding rule: approve funding if and only if $\mu \geq 0.40$. We can formulate this as a hypothesis test:

- **Null hypothesis** H_0 : $\mu < 0.40$ (food insecurity does not warrant funding)
- **Alternative hypothesis** H_1 : $\mu \geq 0.40$ (food insecurity warrants funding)

Suppose we draw a random sample and observe $\bar{Y} = 0.75$. Should we approve funding?

4.2 What is a P-value?

The p-value answers this question: **If the null hypothesis were true (meaning $\mu < 0.40$), how likely would we be to observe data as extreme as what we actually observed (or more extreme)?**

Formally:

$$\text{p-value} = P(\bar{Y} \geq 0.75 \mid H_0 \text{ is true}) \quad (16)$$

To calculate this, we need to specify what " H_0 is true" means precisely. Let's take the most favorable case for the null: $\mu = 0.40$ exactly (the boundary case).

4.3 Calculating the P-value: Method 1 (Enumeration)

Under the null hypothesis $\mu = 0.40$, which samples would give us $\bar{Y} \geq 0.75$? Looking at our complete list:

- Samples with $\bar{Y} = 0.75$: #17, #18, #19, #20 (4 samples)
- Samples with $\bar{Y} = 1.00$: #21 (1 sample)

Total: 5 samples out of 21.

Therefore:

$$\text{p-value} = \frac{5}{21} \approx 0.238 = 23.8\% \quad (17)$$

4.4 Interpreting the P-value

What does $p = 0.238$ mean?

Correct interpretation: If the null hypothesis were true (if μ were really less than 0.40), there would be a 23.8% chance of observing a sample mean as large as 0.75 or larger, purely due to random sampling variability.

Incorrect interpretations (common mistakes):

- ✗ "There's a 23.8% probability that the null hypothesis is true" — NO! The p-value is not $P(H_0 \mid \text{data})$
- ✗ "There's a 23.8% probability we made a mistake" — NO! This confuses p-values with error rates
- ✗ "The effect size is small because $p > 0.05$ " — NO! P-values depend on both effect size and sample size

Decision rule: Using the conventional threshold $\alpha = 0.05$:

- If $p < 0.05$: reject H_0 , conclude evidence favors funding
- If $p \geq 0.05$: fail to reject H_0 , insufficient evidence for funding

In our case, $p = 0.238 > 0.05$, so we fail to reject the null hypothesis. Despite observing $\bar{Y} = 0.75$ —well above 0.40—we cannot definitively conclude that funding is warranted.

Why? Because with such a small sample ($n = 2$), sampling variability is huge. Even if the true mean were below 0.40, we could easily get a sample mean of 0.75 by chance.

4.5 The Logic of Hypothesis Testing

This example illustrates the **logic of falsification** in hypothesis testing:

1. We start by assuming the null hypothesis is true
2. We ask: "How compatible are the observed data with this assumption?"
3. If the data are highly incompatible (small p-value), we reject the null
4. If the data are reasonably compatible (large p-value), we fail to reject the null

Crucially, **failing to reject H_0 is not the same as accepting H_0** . A large p-value might mean:

- The null hypothesis is actually true, OR
- The null hypothesis is false, but our sample is too small to detect the difference

This is why we say "fail to reject" rather than "accept" the null hypothesis.

5 Why Sample Size Matters

5.1 Increasing Sample Size

Let's consider what would happen if we could survey more students. Suppose we could survey $n = 5$ students instead of $n = 2$.

The number of possible samples would be:

$$\binom{7}{5} = \frac{7!}{5!(7-5)!} = 21 \quad (18)$$

Interestingly, this is the same number! (This is because $\binom{7}{5} = \binom{7}{2}$ by symmetry.)

However, the **sampling distribution would be very different**. With $n = 5$, the standard error would be:

$$SE(\bar{Y}) = \sqrt{\frac{S^2}{n} \left(1 - \frac{n}{N}\right)} = \sqrt{\frac{0.2024}{5} \left(1 - \frac{5}{7}\right)} \approx 0.120 \quad (19)$$

Compare this to $SE = 0.318$ when $n = 2$. The standard error decreased by more than half! This means sampling variability is much smaller with a larger sample.

5.2 The General Principle

As sample size increases:

- Standard error decreases (proportional to $1/\sqrt{n}$)
- The sampling distribution becomes more concentrated around μ
- Individual sample means are more likely to be close to the true population mean
- P-values become more sensitive to true effects (statistical power increases)

This is why researchers always want larger samples: not because bias goes away (it doesn't), but because **random sampling error becomes smaller and more predictable**.

5.3 The Fundamental Trade-off

However, there's always a trade-off:

- **Larger samples:** smaller standard errors, more precise estimates, but higher cost
- **Smaller samples:** larger standard errors, less precise estimates, but lower cost

The key is to choose a sample size that balances precision and feasibility. This is the subject of **power analysis** and **sample size planning**, which you'll encounter in more advanced courses.

6 When Random Sampling Fails

6.1 What Makes an Estimator Biased?

So far we've assumed random sampling. But what if the sampling mechanism is not random? What if certain types of people are systematically over- or under-represented?

Sampling bias occurs when the expected value of the sample mean differs from the true population mean:

$$\text{Bias} = \mathbb{E}[\bar{Y}] - \mu \neq 0 \quad (20)$$

Crucially, bias does not decrease with sample size. You can have a huge sample and still have severe bias if the sampling mechanism is flawed.

6.2 Example: Convenience Sampling

Suppose instead of random sampling, we use **convenience sampling**: we survey the first 2 students we encounter in the Student Union Building.

Who hangs out in the SUB? Probably students who are *not* worried about food (because they have money to buy food there). This would systematically *under-represent* food-insecure students.

Let's say the convenience sample tends to select from students $\{1, 2, 3, 4, 5\}$ (the less food-insecure ones) and almost never selects students $\{6, 7\}$ (the always-worried ones).

If we enumerate all possible samples from this biased mechanism, the expected value might be:

$$\mathbb{E}[\bar{Y}] \approx 0.20 < \mu = 0.43 \quad (21)$$

This is **negative bias**: our sampling method systematically underestimates the true population mean.

6.3 Why Bias is Fatal

Here's the critical difference between bias and random error:

Feature	Random Error	Bias
Expected value	$\mathbb{E}[\bar{Y}] = \mu$	$\mathbb{E}[\bar{Y}] \neq \mu$
Effect of larger sample	Decreases ($\propto 1/\sqrt{n}$)	No change
Can be quantified?	Yes (standard error)	Often unknown
Can be fixed?	No (inherent to sampling)	Yes (improve design)

Table 3: Random sampling error vs. systematic bias

Random error is unavoidable but manageable:

- It averages out to zero in the long run
- It decreases with larger samples
- We can quantify it using standard errors
- We can account for it using confidence intervals and p-values

Bias is a design flaw:

- It does not average out—it's systematic
- It persists no matter how large the sample
- We often cannot quantify it (because we don't know the population truth)
- No statistical method can "fix" bias after data collection

6.4 Sources of Sampling Bias

Common sources of bias in real research:

1. Convenience sampling

- Selecting whoever is easiest to reach
- Example: surveying only students in a particular building

- Result: systematically misses certain subgroups

2. Voluntary response bias

- Only people who choose to respond are included
- Example: online polls, call-in surveys
- Result: over-represents people with strong opinions

3. Non-response bias

- Even with random sampling, if certain types of people don't respond
- Example: if food-secure students are more likely to complete the survey
- Result: sample composition differs from population

4. Survivorship bias

- Only observing units that "survived" some selection process
- Example: surveying only current students (misses those who dropped out due to financial stress)
- Result: systematically excludes relevant subgroup

6.5 The Hierarchy of Evidence Quality

From worst to best:

1. **Convenience sample, no randomization:** High bias, no randomization
2. **Convenience sample, with randomization:** High sampling bias, but causal inference possible within sample
3. **Random sample, no randomization:** Generalizable associations, but not causal
4. **Random sample, with randomization:** Gold standard—both generalizable and causal

The take-home message: **No amount of sophisticated statistical analysis can compensate for poor study design.** Garbage in, garbage out.

6.6 Final Thoughts

The core challenge of statistical inference is inferring the whole (population) from the part (sample). Random sampling provides us with a powerful but limited tool: it guarantees unbiasedness and quantifiable uncertainty, but doesn't eliminate error and cannot correct bias. The framework you've learned today—from concrete numerical calculations to abstract conceptual understanding—forms the foundation for all of statistical inference. When you see p-values in regression output, confidence intervals in published research, or claims about "statistical significance" in the news, you now understand the logic behind them: it all comes from an imagined sampling distribution, measuring how rare the current data would be if the null hypothesis were true. Statistical tools are useful, but they're not magic—they rest on random sampling assumptions, can only quantify random error, and cannot substitute for good study design and critical thinking.

A Complete Sample List

Table 4: Complete enumeration: All samples and their means

Sample #	Students (i, j)	Values (Y_i, Y_j)	Mean $\bar{Y}$
1	(1,2)	(0,0)	0.00
2	(1,3)	(0,0)	0.00
3	(1,4)	(0,0.5)	0.25
4	(1,5)	(0,0.5)	0.25
5	(1,6)	(0,1)	0.50
6	(1,7)	(0,1)	0.50
7	(2,3)	(0,0)	0.00
8	(2,4)	(0,0.5)	0.25
9	(2,5)	(0,0.5)	0.25
10	(2,6)	(0,1)	0.50
11	(2,7)	(0,1)	0.50
12	(3,4)	(0,0.5)	0.25
13	(3,5)	(0,0.5)	0.25
14	(3,6)	(0,1)	0.50
15	(3,7)	(0,1)	0.50
16	(4,5)	(0.5,0.5)	0.50
17	(4,6)	(0.5,1)	0.75
18	(4,7)	(0.5,1)	0.75
19	(5,6)	(0.5,1)	0.75
20	(5,7)	(0.5,1)	0.75
21	(6,7)	(1,1)	1.00

B Key Formulas

Quantity	Formula
Population mean	$\mu = \frac{1}{N} \sum_{i=1}^N Y_i$
Population variance	$S^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i - \mu)^2$
Sample mean	$\bar{Y} = \frac{1}{n} \sum_{j=1}^n Y_{(j)}$
Unbiasedness	$\mathbb{E}[\bar{Y}] = \mu$
Variance (with FPC)	$\text{Var}(\bar{Y}) = \frac{S^2}{n} \left(1 - \frac{n}{N}\right)$
Standard error	$\text{SE}(\bar{Y}) = \sqrt{\text{Var}(\bar{Y})}$
95% Margin of error	$\text{MOE}_{95\%} = 1.96 \times \text{SE}(\bar{Y})$
Finite population correction	$\text{FPC} = 1 - \frac{n}{N}$
Bias	$\text{Bias} = \mathbb{E}[\bar{Y}] - \mu$

Table 5: Key formulas summary for finite population simple random sampling

C Proof of Unbiasedness

Theorem: Under simple random sampling without replacement, the sample mean $\bar{Y}$ is an unbiased estimator of the population mean μ .

Proof:

Define indicator variables $S_i \in \{0, 1\}$, where $S_i = 1$ if and only if unit i is selected into the sample. Under SRSWOR:

$$\mathbb{E}[S_i] = P(S_i = 1) = \frac{n}{N} \quad (22)$$

This is because of the N units, n are selected, and each unit has equal probability of selection.

The sample mean can be written as:

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^N S_i Y_i \quad (23)$$

(Only selected units contribute to the sum)

Taking expectations:

$$\mathbb{E}[\bar{Y}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^N S_i Y_i\right] \quad (24)$$

$$= \frac{1}{n} \sum_{i=1}^N \mathbb{E}[S_i Y_i] \quad (25)$$

$$= \frac{1}{n} \sum_{i=1}^N \mathbb{E}[S_i] \cdot Y_i \quad (\text{since } Y_i \text{ is fixed population value}) \quad (26)$$

$$= \frac{1}{n} \sum_{i=1}^N \frac{n}{N} Y_i \quad (27)$$

$$= \frac{1}{N} \sum_{i=1}^N Y_i \quad (28)$$

$$= \mu \quad \checkmark \quad (29)$$

Therefore $\bar{Y}$ is an unbiased estimator of μ . $\square$

D Concept Glossary

Population	Complete set of units we care about; parameters denoted by Greek letters (e.g., μ, β)
Sample	Subset we actually observe; statistics denoted by Latin letters (e.g., $\bar{Y}, \hat{\beta}$)
Parameter	Numerical characteristic of population (e.g., population mean μ); fixed but unknown
Statistic	Numerical characteristic of sample (e.g., sample mean $\bar{Y}$); observable and varies across samples
Sampling Error	Difference between statistic and parameter, $\bar{Y} - \mu$; random and unavoidable
Sampling Distribution	Probability distribution of a statistic across all possible samples; theoretical distribution
Unbiasedness	$\mathbb{E}[\bar{Y}] = \mu$; correct on average, but doesn't guarantee individual sample accuracy
Standard Error	Standard deviation of a statistic; measures typical size of sampling error

P-value	Probability of observing current data or more extreme data if null hypothesis is true; NOT the probability hypothesis is true
Confidence Interval	An interval estimation procedure with specified long-run coverage rate (e.g., 95%)
Finite Population Correction (FPC)	Factor $(1 - n/N)$ that reduces variance when sampling fraction n/N is non-negligible
Sampling Bias	Systematic over- or under-representation such that $\mathbb{E}[\bar{Y}] \neq \mu$; fatal design flaw
Significance Level	Denoted α , typically 0.05; p-value threshold for rejecting null hypothesis
Type I Error	Incorrectly rejecting true null hypothesis; probability is α
Type II Error	Failing to reject false null hypothesis; probability is β (not regression coefficient!)
Power	Probability of correctly rejecting false null hypothesis; equals $1 - \beta$